

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 1.4224400602169534, median 1.227150628134017, std: 0.9260537445049435
Reprojection error (cam1): mean 1.4378863712293706, median 1.275151785226356, std: 0.8893535601249887
Gyroscope error (imu0): mean 2.3758071756297165, median 2.005063675118185, std: 1.4894402957076653
Accelerometer error (imu0): mean 0.6105151233230305, median 0.5392299591688425, std: 0.3501333981444204

Residuals

Reprojection error (cam0) [px]: mean 1.4224400602169534, median 1.227150628134017, std: 0.9260537445049435
Reprojection error (cam1) [px]: mean 1.4378863712293706, median 1.275151785226356, std: 0.8893535601249887
Gyroscope error (imu0) [rad/s]: mean 0.023758071756297162, median 0.020050636751181853, std: 0.014894402957076653
Accelerometer error (imu0) [m/s^2]: mean 0.24420604932921214, median 0.215691983667537, std: 0.14005335925776816

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.00573653 0.03121121 0.99949635 0.0120595]
 [-0.01649305 0.99937979 -0.03111291 0.1689564]
 [-0.99984752 -0.01630626 0.00624774 -0.12891039]
 [0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[0.00573653 -0.01649305 -0.99984752 -0.12617331]
 [0.03121121 0.99937979 -0.01630626 -0.17133005]
 [0.99949635 -0.03111291 0.00624774 -0.0059913]
 [0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.00021261455265482617

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.0092135 0.02573979 0.99962622 -0.03891404]
[-0.01525727 0.9995559 -0.02559735 0.16876483]
[-0.99984115 -0.01501572 0.00960213 -0.12907813]
[0. 0. 0. 1.]]

T_ic: (cam1 to imu0):
[[0.0092135 -0.01525727 -0.99984115 -0.1261242]
[0.02573979 0.9995559 -0.01501572 -0.16962645]
[0.99962622 -0.02559735 0.00960213 0.04445886]
[0. 0. 0. 1.]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.00023733691095592633

Baselines:

Baseline (cam0 to cam1):
[[0.99997898 -0.00552942 -0.00338641 -0.0504756]
[0.00552537 0.99998401 -0.00120401 -0.00041071]
[0.00339301 0.00118527 0.99999354 -0.00040975]
[0. 0. 0. 1.]]
baseline norm: 0.050478933675274874 [m]

Gravity vector in target coords: [m/s^2]
[0.28866564 -9.80215509 -0.05339082]

Calibration configuration
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cam0

Camera model: pinhole
Focal length: [757.1900150812725, 767.3209550918491]
Principal point: [623.6126208257, 384.12750918399144]
Distortion model: radtan
Distortion coefficients: [-0.01659061452659088, -0.02693348950149743, -0.0038990320324666884,
0.00139549935042361]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.02 [m]
Spacing 0.006 [m]

cam1

Camera model: pinhole
Focal length: [755.8431637852096, 767.2329918669442]
Principal point: [630.165403751185, 364.9061849564123]
Distortion model: radtan
Distortion coefficients: [-0.019721425278644445, -0.021337065708098103, -0.0034344689446923877,
0.0016636586453975452]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.02 [m]
Spacing 0.006 [m]

IMU configuration

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IMU0:

Model: calibrated
Update rate: 400

Accelerometer:

Noise density: 0.02

Noise density (discrete): 0.39999999999999997

Random walk: 0.001

Gyroscope:

Noise density: 0.0005

Noise density (discrete): 0.01

Random walk: 0.0001

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

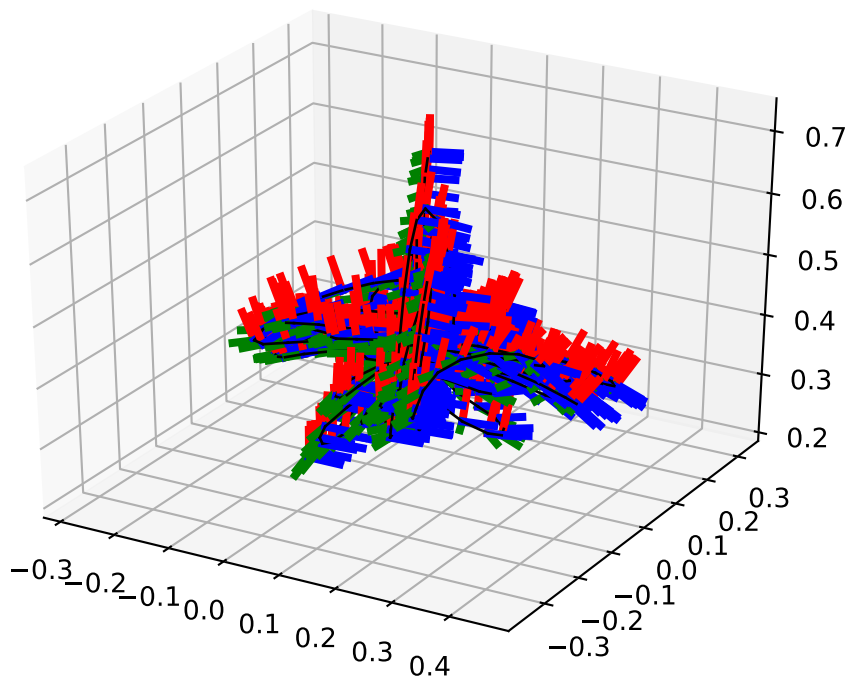
[0. 1. 0. 0.]

[0. 0. 1. 0.]

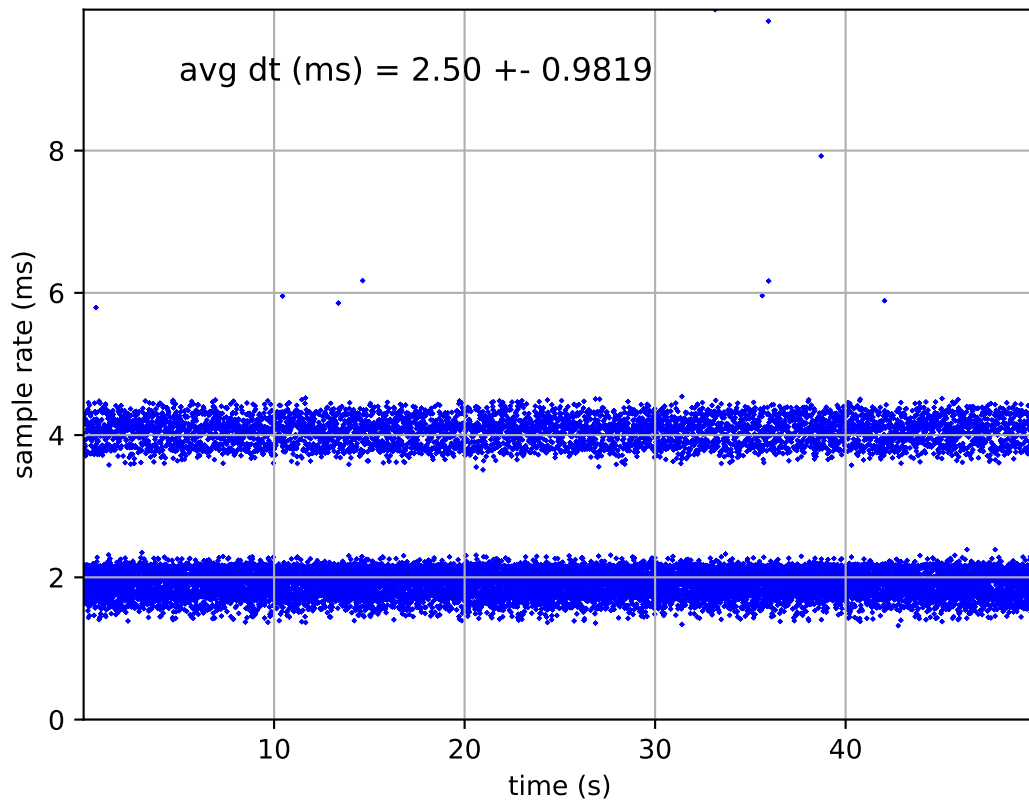
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

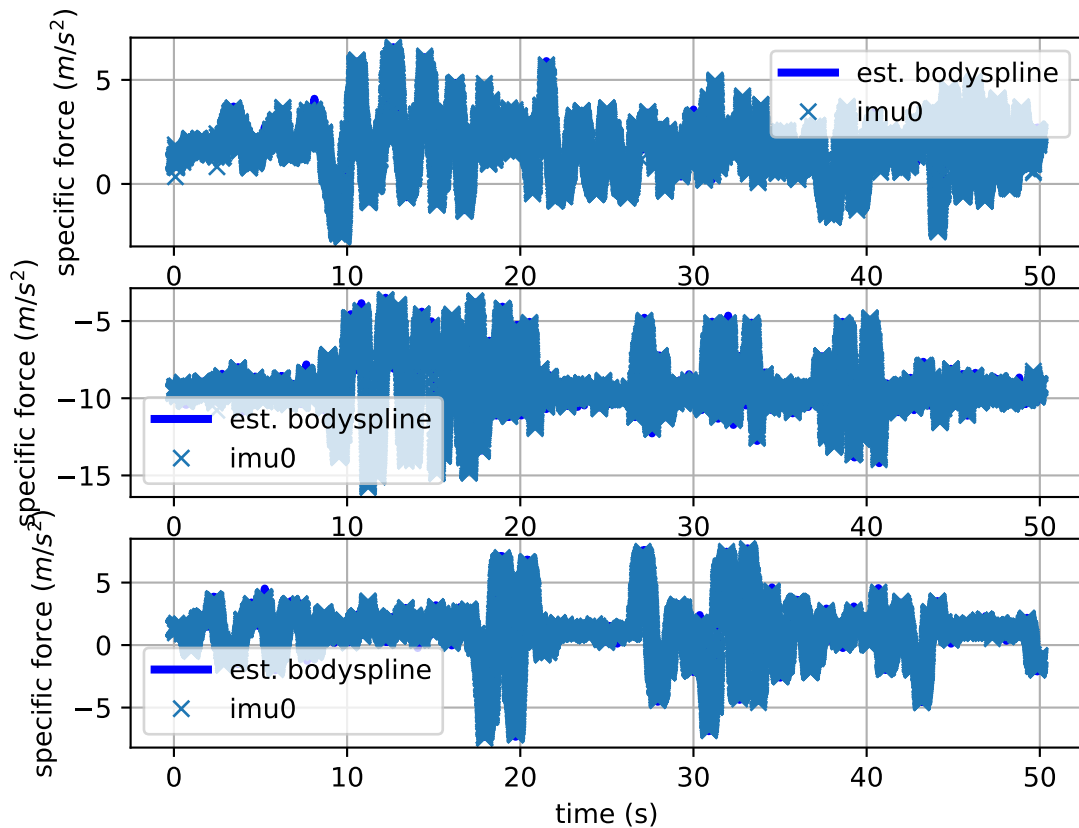
imu0: estimated poses



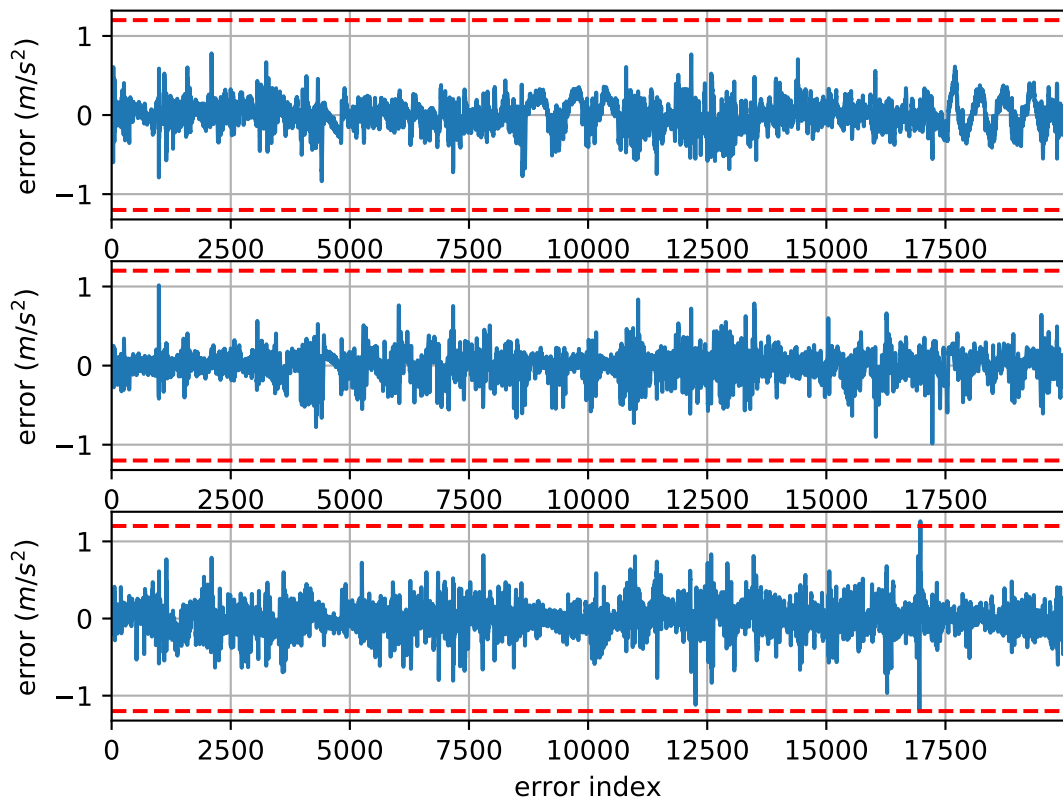
imu0: sample inertial rate



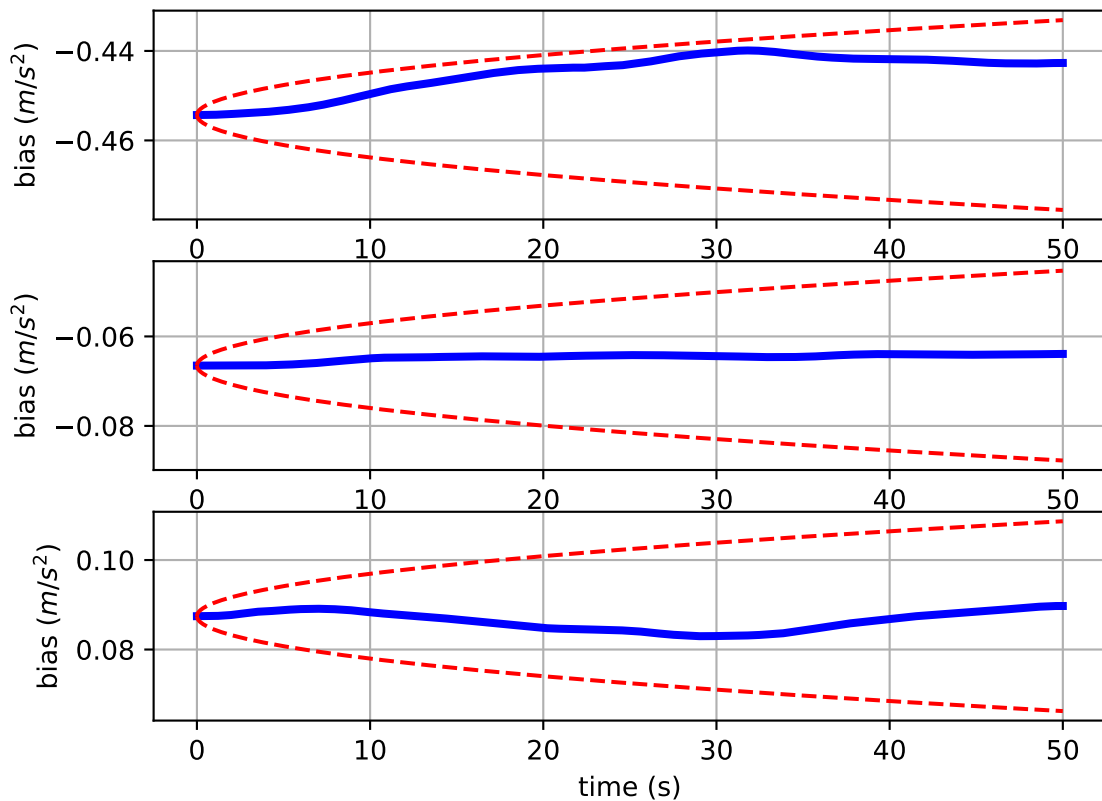
Comparison of predicted and measured specific force (imu0 frame)



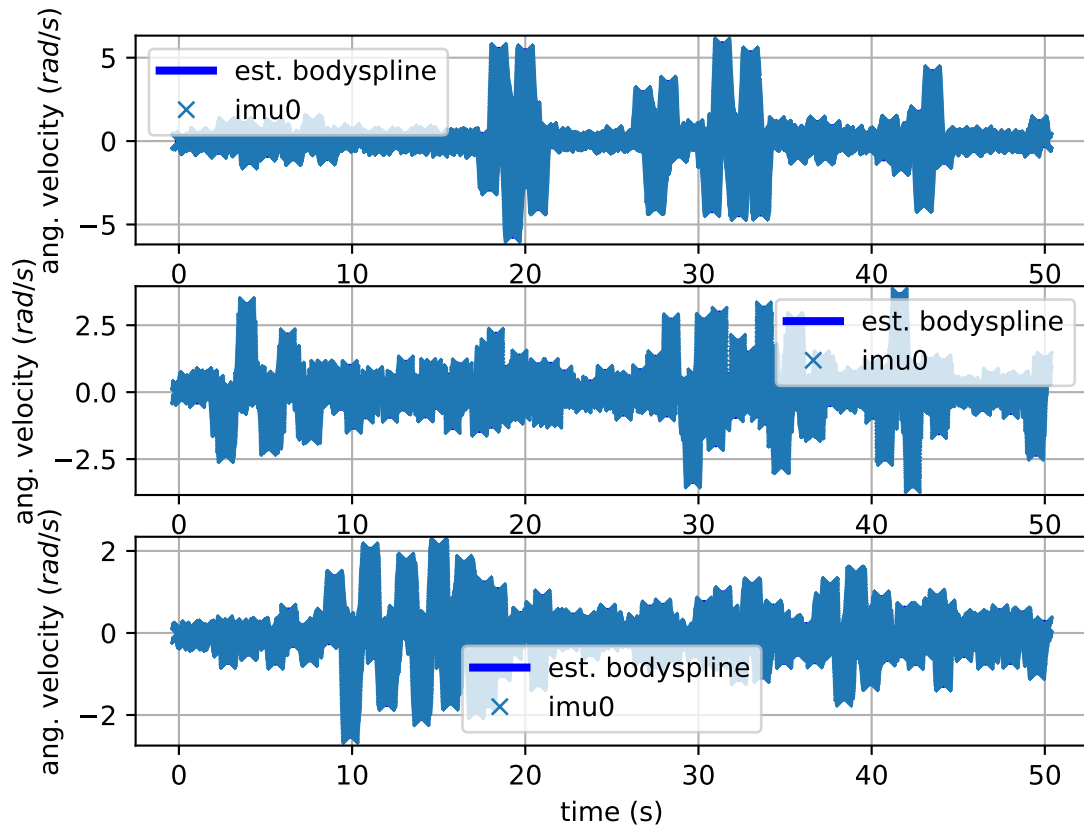
imu0: acceleration error



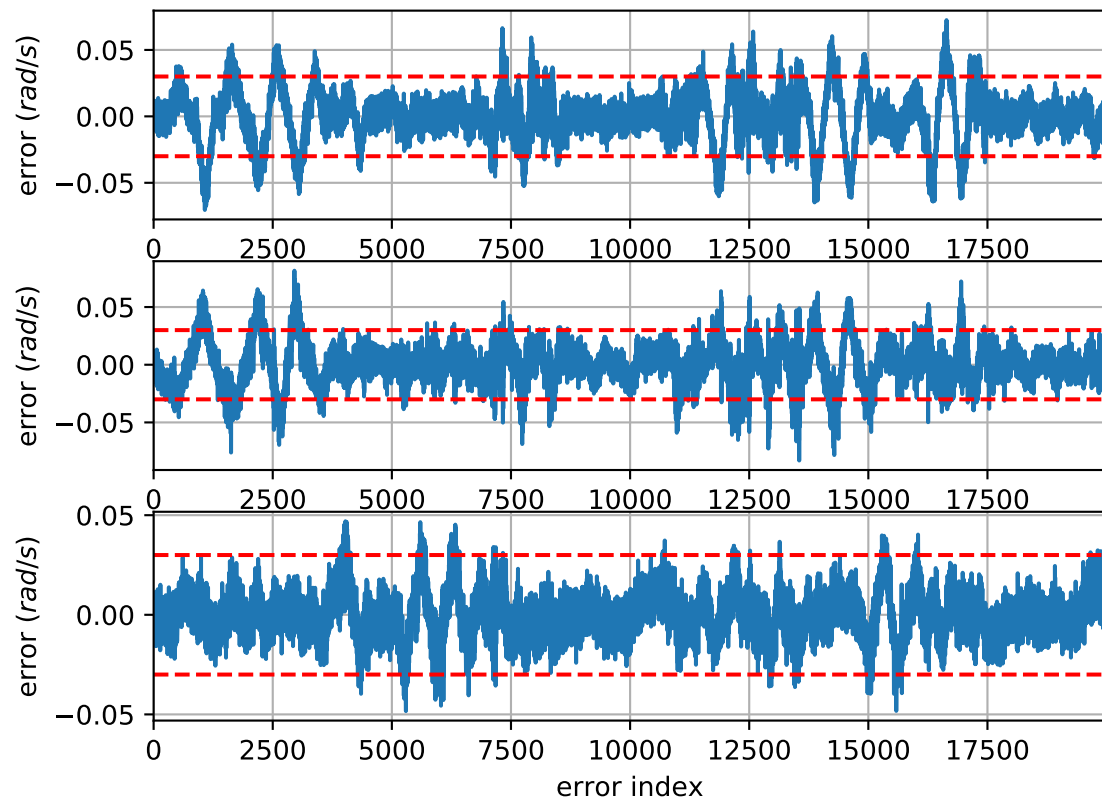
imu0: estimated accelerometer bias (imu frame)



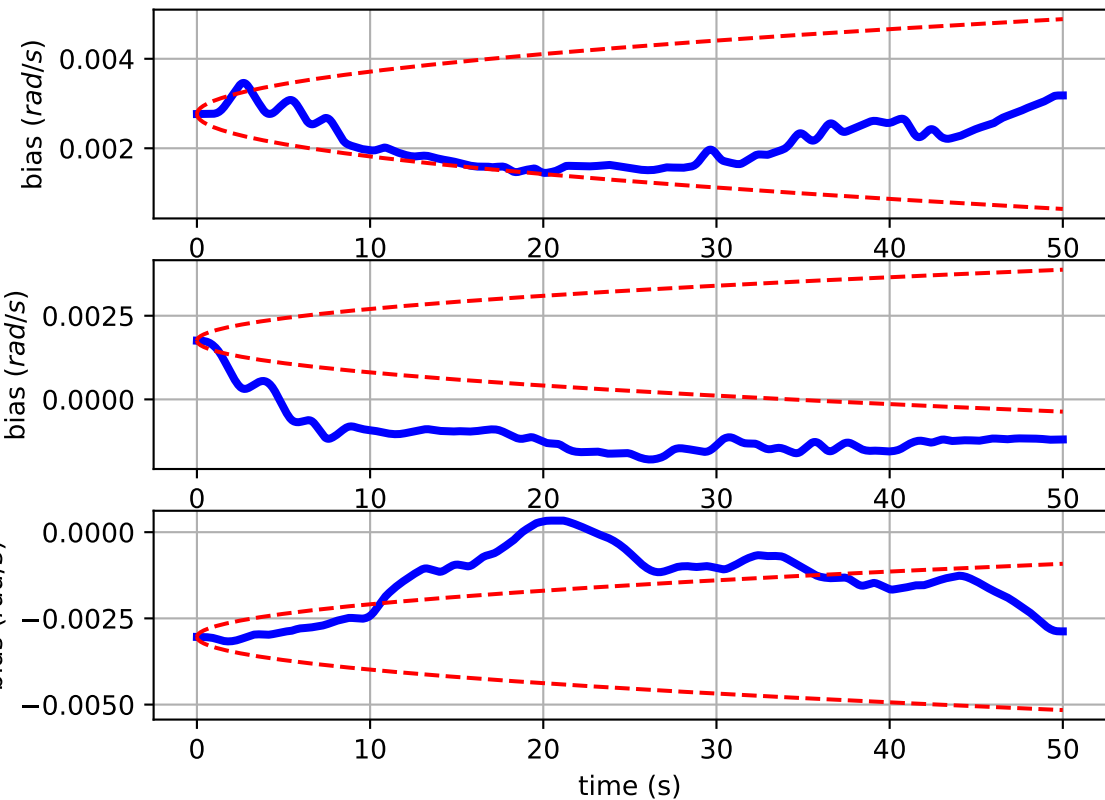
Comparison of predicted and measured angular velocities (body frame)



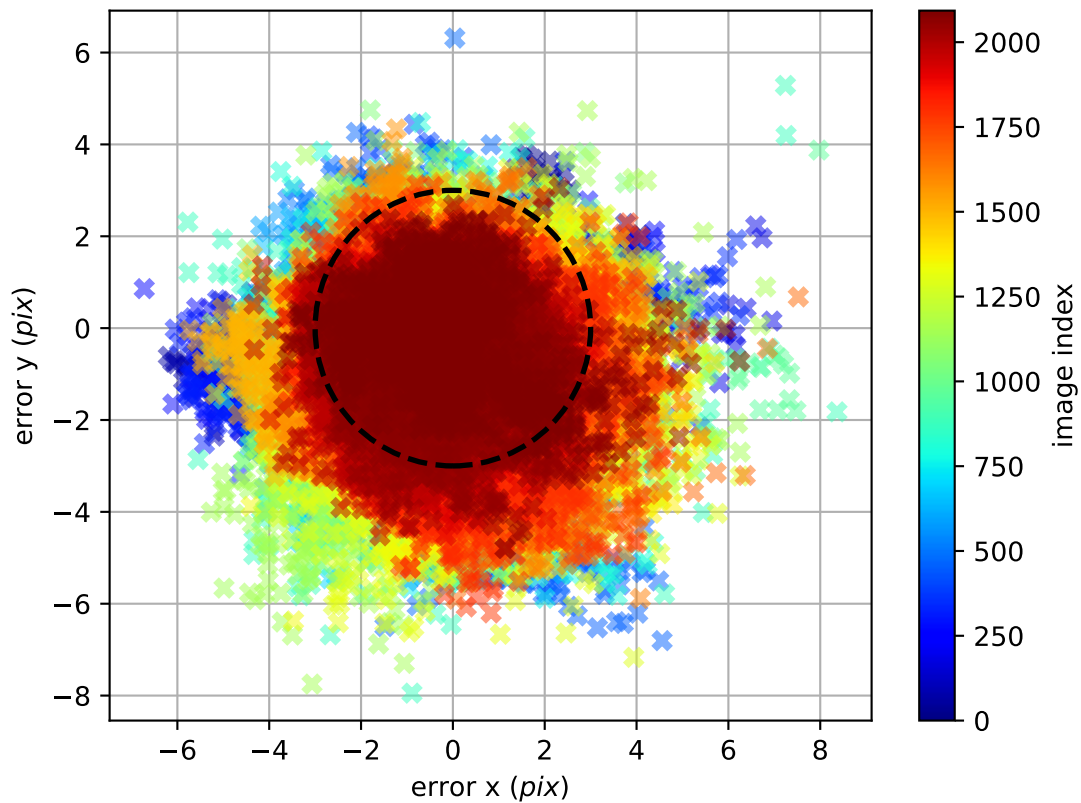
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

